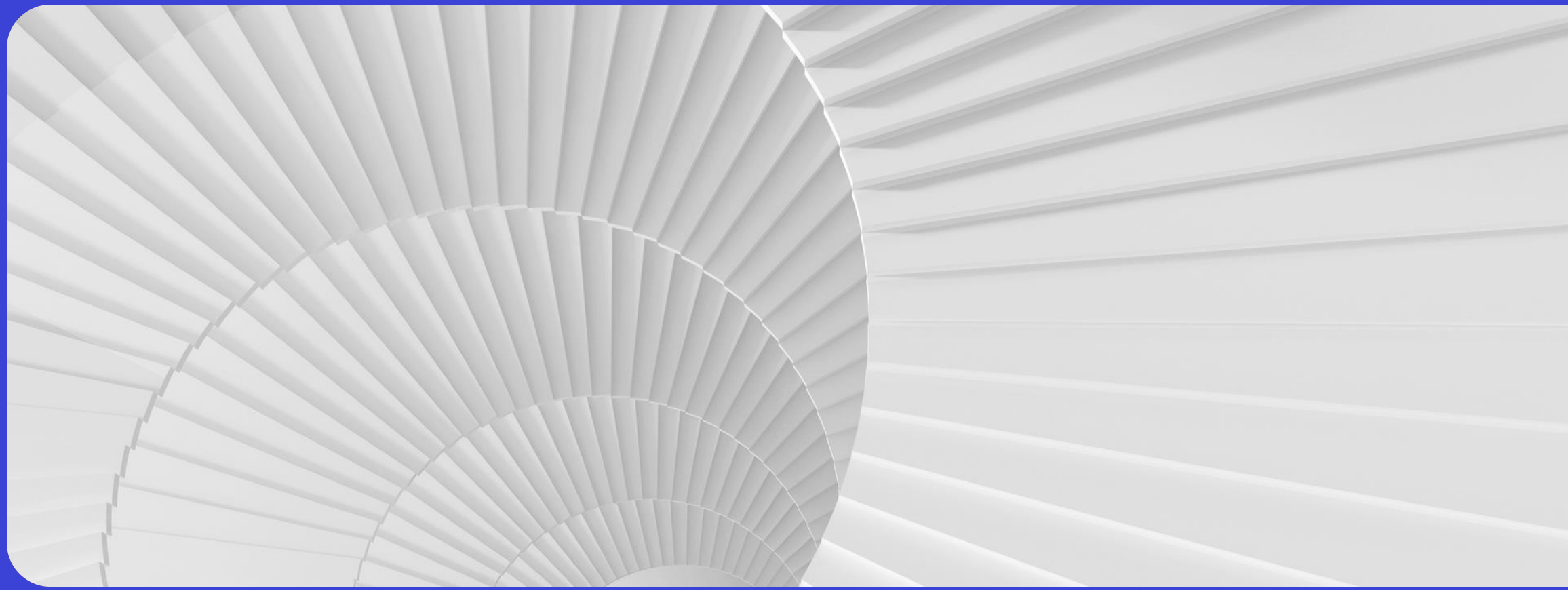


Vehicle Sales & Inventory Analysis

Presented by Ganesh Singh

1.Executive Summary

Business Problem: Car Dekho, a leading automotive dealership platform, lacks a centralized data analytics system to understand sales trends, inventory performance, and customer fuel preferences. The management team relies on intuition rather than data-driven insights for inventory planning and marketing strategy.

Proposed Solution: Develop a SQL-based analytics system to analyze historical sales data, identify trends, and provide actionable insights for business decision-making.

Expected Outcome: Data-driven inventory management, optimized marketing spend, and improved sales forecasting accuracy.

2. Business Background

2.1 About the Company

Car Dekho is an automotive marketplace connecting buyers with car sellers. They manage inventory across multiple dealerships, offering various fuel types (Petrol, Diesel, CNG) across different car models and price ranges.

2.2 Current Challenges

Challenge	Business Impact
Unclear Sales Trends	Cannot predict peak seasons or declining demand
Poor Inventory Management	Overstock of slow-moving models, understock of popular ones
No Fuel Type Analysis	Don't know which fuel types customers prefer year-over-year
No Year-over-Year Comparison	Cannot measure growth or decline accurately
Reactive Marketing	Marketing campaigns launched without data support

3. Stakeholders & Their Needs

Stakeholder		Key Questions	What They Need
CEO / Management		• Which years performed best? • Is the business growing? • Which fuel types should we focus on?	High-level dashboard with key metrics and trends
Inventory Manager		• Which models sell most? • When to increase/decrease inventory? • Which fuel types are declining?	Yearly sales trends, fuel type demand patterns
Marketing Head		• Which years had highest sales? • Which fuel types to promote? • When to run promotions?	Market share analysis, peak season identification
Sales Team		• Which cars are most popular? • What are customers preferring? • Which years had best response?	Top performing models, customer preference trends
Finance / Investors		• Is business growing consistently? • Which segments are profitable? • Future growth potential?	Growth rate analysis, market share, ROI projections

4. Business Objectives

4.1 Primary Objectives

#	Objective	Success Metric
1	Understand overall sales volume	Total cars sold across all years
2	Identify peak sales years	Years with 1000+ car sales
3	Analyze fuel type demand	Distribution of Petrol, Diesel, CNG vehicles
4	Track year-over-year growth	Percentage growth/decline year by year
5	Enable data-driven inventory planning	Recommendations based on trend analysis

4.2 Secondary Objectives

#	Objective	Success Metric
6	Identify market share of fuel types	Percentage share of each fuel type
7	Forecast future demand	Trend-based predictions for next 2-3 years
8	Optimize marketing spend	Targeted campaigns during high-demand periods
9	Support investor reporting	Clear visualizations and metrics

5. Key Business Questions (KPIs)

5.1 Sales Performance

KPI	Business Question
Total Sales Volume	How many total cars are in our database?
Yearly Sales Trend	How has sales performed year over year?
Top Sales Years	Which years had the highest sales?
Growth Rate	What is our year-over-year growth percentage?

5.2 Fuel Type Analysis

KPI	Business Question
Fuel Distribution	What is the market share of Petrol vs Diesel vs CNG?
Fuel Preference by Year	Are customers shifting from Diesel to Petrol?
New Fuel Adoption	When were new fuel types introduced in our inventory?
Fuel Consistency	Which fuel type has the most stable demand?

5.4 Financial & Strategic

KPI	Business Question
Cumulative Sales	What is our total sales over the entire period?
Sales Targets	Which years exceeded 1000 units?
Growth Consistency	Have we shown consistent growth over 3+ years?
Market Leadership	Which fuel type dominates our sales?

5.3 Inventory & Operations

KPI	Business Question
High-Volume Years	Which years should we increase inventory?
Low-Volume Years	Which years require promotional campaigns?
Fuel Availability	Should we stock more Petrol or Diesel vehicles?
Market Shifts	When did major fuel preference shifts occur?

6. Data Requirements

6.1 Required Data Elements

Column Name	Data Type	Description	Business Use
name	VARCHAR	Car model name	Product analysis
year	INT	Manufacturing/Sale year	Time trend analysis
selling_price	INT	Sale price in INR	Revenue analysis
km_driven	INT	Distance driven	Usage pattern analysis
fuel	VARCHAR	Fuel type (Petrol/Diesel/CNG)	Fuel preference analysis
seller_type	VARCHAR	Individual/Dealer	Channel performance
transmission	VARCHAR	Manual/Automatic	Feature preference
owner	VARCHAR	First/Second/Third	Ownership trends

6.2 Data Sources

> Internal sales database

- Dealership records
- Customer purchase history

6.3 Data Quality Requirements

- Complete records with no null values
- Consistent fuel type naming conventions
- Valid year range (e.g., 2000-2024)
- Positive price and distance values

Budget allocation & growth forecast

Category	Budget allocation	Projected growth	ROI expectation
R&D investment	$75M	+20% annually	High
Marketing	$50M	+15% annually	Moderate
Operations	$40M	+5% annually	Stable
Tech upgrades	$30M	+25% annually	High
Market expansion	$80M	+30% annually	High
Cash reserves	$25M	N/A	Essential

“The first one gets the oyster, the second gets the shell.”

Andrew Carnegie

Strategies for sustainable growth

High-return opportunities

Targeted investments support long-term expansion. By focusing on high-return opportunities, we can maximize our growth potential and achieve sustainable success.

Technology integration

Leveraging AI and automation to drive financial efficiency. By integrating advanced technologies into our operations, we can streamline processes, reduce costs, and enhance productivity.

Market expansion

Entering new territories to maximize profitability. Expanding into new markets allows us to diversify our revenue streams and reduce our reliance on existing markets.

Risk management

We implement strategies to hedge against financial uncertainties. By proactively managing risks, we can safeguard our business against potential threats and ensure long-term stability.

Moving forward with confidence

Financial stability

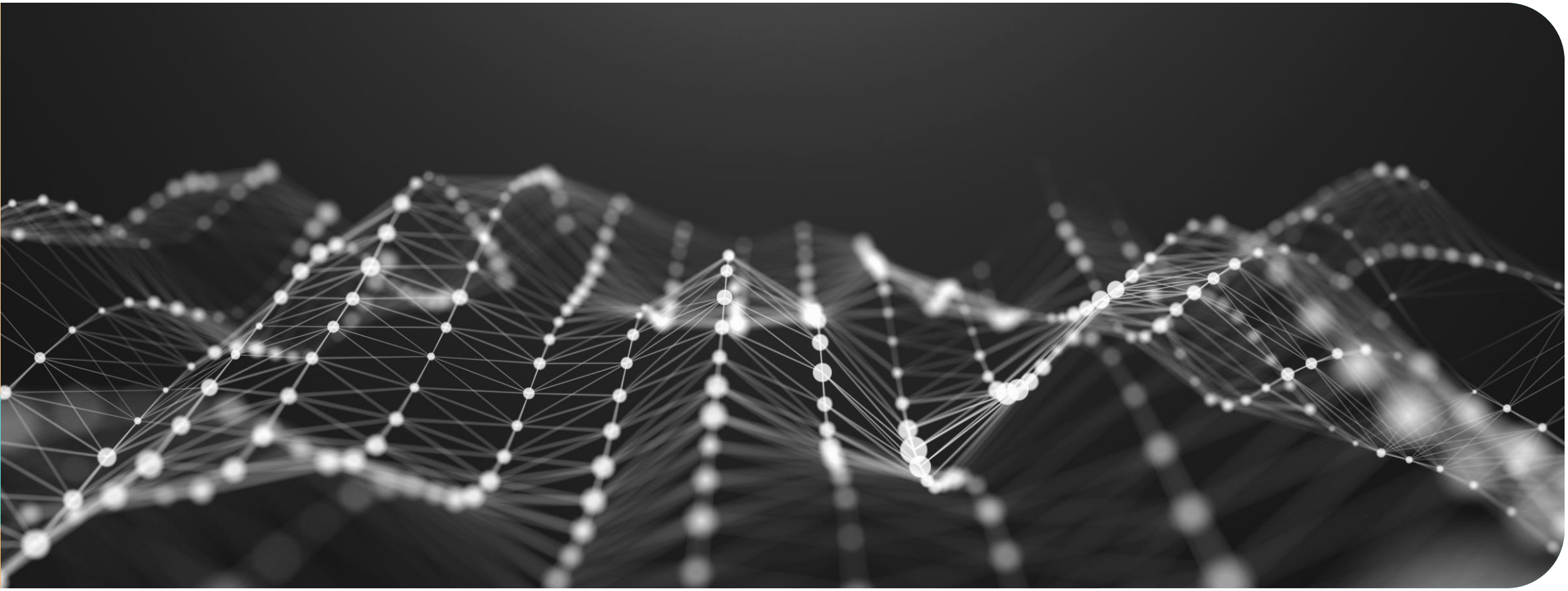
We remain committed to strong financial management.

Strategic investments

Our next steps focus on long-term value creation.

Sustained growth

We're positioned for scalable expansion in the coming years.



Thank you